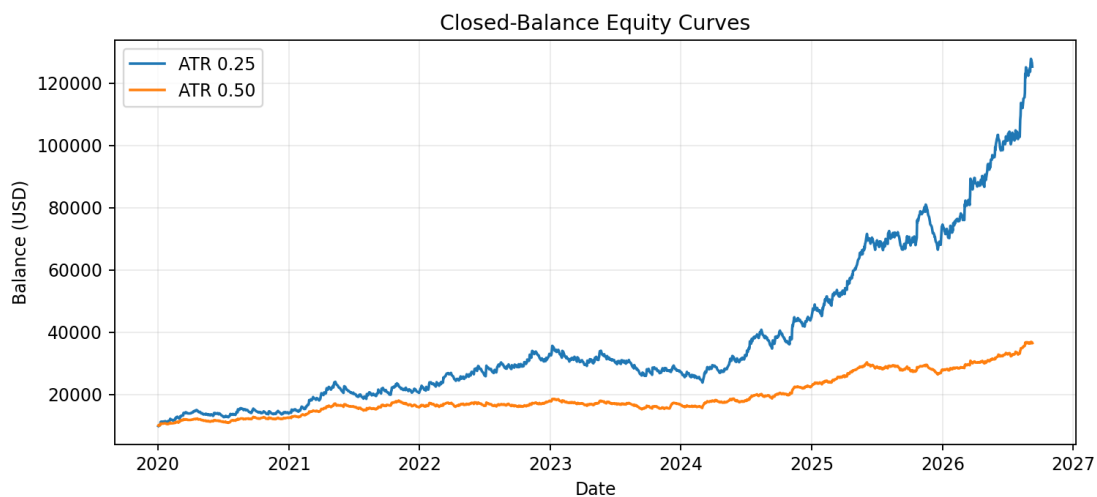


ORB GOLD - DETAILED RESEARCH REPORT

ATR Stop 0.25 vs ATR Stop 0.50 - No Opening-Range Filter

Instrument: XAUUSD Test period: 2020-01-01 to 2026-09-09 Initial balance: \$10,000
Core session: 03:05-06:05 Close: 18:55 ATR entry buffer: 0.05 Risk: 1% per trade

This report independently reconstructs all 1,697 completed trades from each uploaded MT5 report, then calculates daily/trade-level, weekly, monthly, yearly, drawdown, concentration, timing and paired-variant statistics. MT5 headline statistics are shown alongside independently calculated closed-balance metrics where useful.



Headline: ATR 0.25 is the clearly stronger growth engine, but it obtains that growth by accepting a much lower win rate, roughly twice the drawdown severity, a 17-loss streak, and greater dependence on large trend days. ATR 0.50 is materially smoother but leaves substantial trend payoff on the table.

1. Executive Summary

Metric	ATR 0.25	ATR 0.50	Interpretation
Final balance	\$125,274.47	\$36,532.67	0.25 compounds much faster
Net profit	\$115,274.47	\$26,532.67	4.35x more net profit for 0.25
Total return	1152.74%	265.33%	Same starting capital
CAGR (independent)	45.95%	21.38%	0.25 roughly doubles annualized growth
Profit factor	1.333	1.248	Both profitable; 0.25 stronger
Win rate	39.01%	49.79%	0.25 trades lower hit rate for larger payoff
Avg win / avg loss	1.86% / -0.93%	0.83% / -0.67%	Normalized to balance at entry
MT5 equity DD relative	33.57% (12,007.62)	18.77% (3,519.18)	0.25 materially more aggressive
Max loss streak	17	9	Psychological/prop-firm relevance
Positive months	59.26%	59.26%	Exactly the same 48/81 positive months

Research conclusion: These are good tests and the strategy has a real-looking breakout payoff structure, not a simple high-win-rate curve fit. The 0.25 ATR stop is the more powerful version, but it should not yet be declared the final production setting. The key next question is whether a middle zone (roughly 0.30-0.40 ATR) or the 0.25 stop at lower account risk can retain most of the CAGR while reducing the 33% closed-balance drawdown and 17-loss streak.

The strongest evidence in favor of the system is persistence across the full 2020-2026 sample, six positive calendar years out of seven, 99% history quality, and 1,697 trades. The main caution is regime sensitivity: 2023 was negative for both versions and the recovery from the January 2023 peak took roughly 1.4-1.5 years.

2. Strategy Configuration and Test Quality

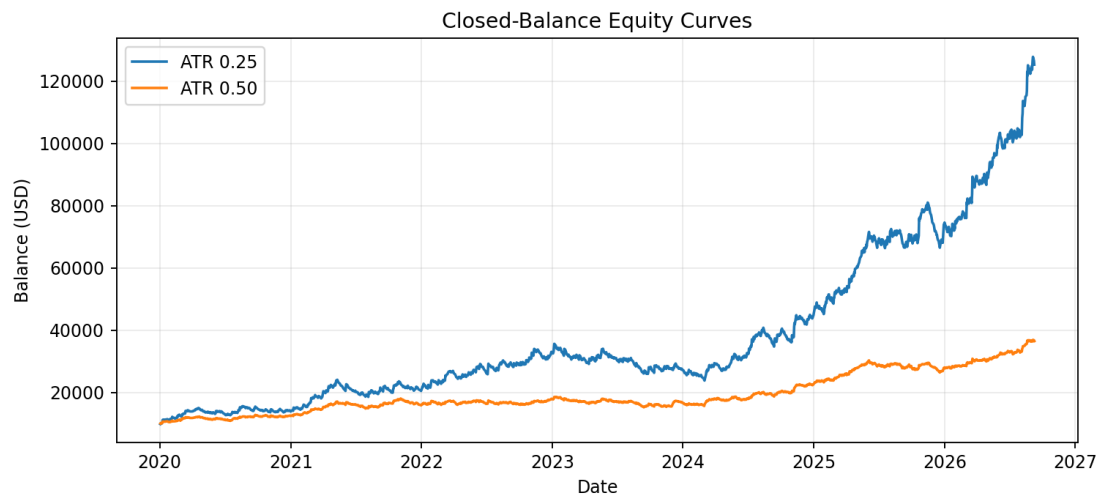
Field	Value
Expert	ORB_V2
Symbol / timeframe	XAUUSD / M15
Test period	2020-01-01 to 2026-09-09
History quality	99%
Bars / ticks	158,082 bars / 282,827,485 ticks
Opening range	03:05 to 06:05 (selected session clock)
Entry cutoff	Disabled
Trade close	18:55
Entry signal timeframe	M5
Entry ATR	ATR(14), buffer multiplier 0.05
Opening-range filters	Price-% filter OFF; ATR range filter OFF
Strong-candle filter	OFF
Directions / daily trade cap	Long + Short enabled; max 1 trade/day
Stop variants	ATR stop multiplier 0.25 vs 0.50
Risk model	1.0% balance risk per trade
Spread filter	Max spread = 0.15 ATR
Slippage setting	20 points
Explicit commission/swap in deals	\$0 / \$0

Important limitation: the reports show zero explicit commission and swap. Bid/ask spread effects are present through execution prices and the EA has a spread filter, but a Raw-spread live account may also charge commission. A cost-sensitivity rerun should therefore be mandatory before live deployment.

Because both variants have the same 1,697 entry events, the stop comparison is unusually clean: the entry signal itself is held constant, while stop distance and risk-based position sizing change the exit path and payoff distribution.

3. Headline Performance - MT5 and Independent Calculations

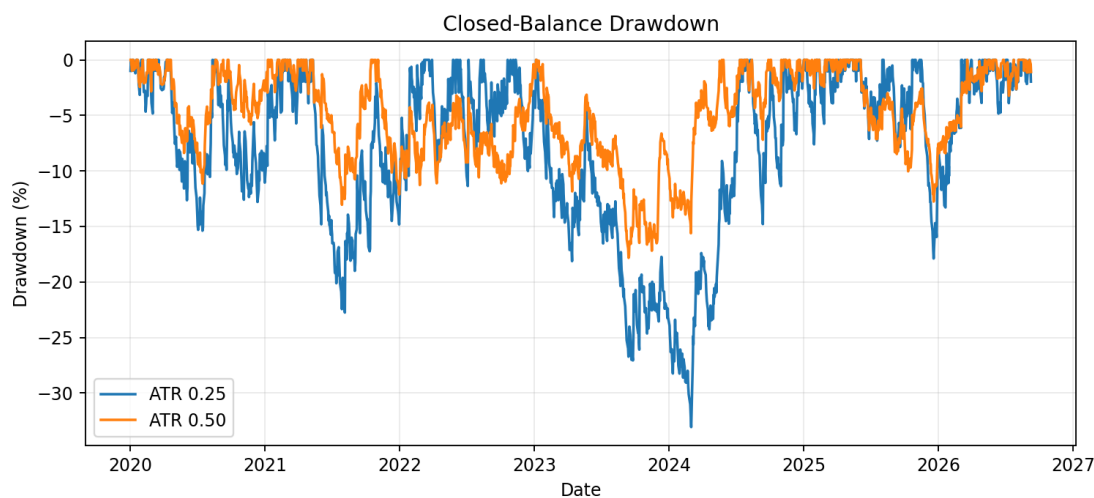
Metric	ATR 0.25	ATR 0.50
Initial deposit	\$10,000.00	\$10,000.00
Final closed balance	\$125,274.47	\$36,532.67
Net profit	\$115,274.47	\$26,532.67
Total return	1152.74%	265.33%
Independent CAGR	45.95%	21.38%
MT5 profit factor	1.333	1.248
MT5 expected payoff	\$67.93	\$15.64
MT5 Sharpe ratio	2.504	2.003
Monthly-return Sharpe (0% RF)	1.60	1.34
Monthly-return Sortino	4.54	3.27
Independent Calmar	1.39	1.20
MT5 recovery factor	6.64	6.52
MT5 LR correlation	0.874	0.905



ATR 0.25 converts the same signal stream into much more convex returns. Its CAGR is about 45.95% versus 21.38% for ATR 0.50. This does not come from a higher hit rate; it comes from larger position size under the fixed 1% risk model and from allowing successful breakouts to produce much larger R-multiples before the 18:55 exit.

4. Drawdown, Recovery and Path Risk

Metric	ATR 0.25	ATR 0.50
MT5 balance DD relative	33.07% (11,793.23)	17.84% (3,319.32)
MT5 equity DD relative	33.57% (12,007.62)	18.77% (3,519.18)
Closed-balance max DD	33.07%	17.84%
Closed DD peak -> trough	2023-01-06 -> 2024-03-01	2023-01-06 -> 2023-09-13
Longest closed-equity recovery	552 days	503 days
Longest recovery window	2023-01-06 -> 2024-07-11	2023-01-06 -> 2024-05-23
Max consecutive losses	17	9
Max consecutive wins	8	14

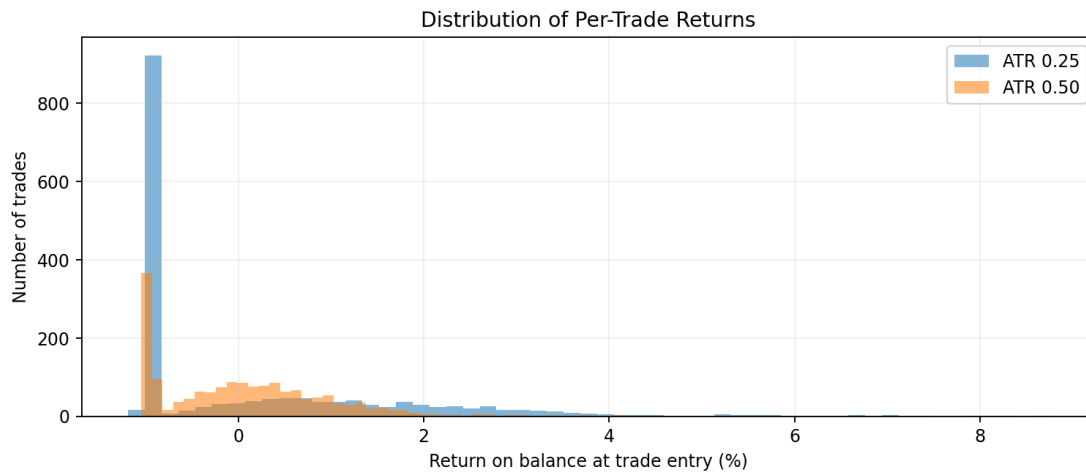


The 2023 drawdown is the critical stress episode. Both variants peak on 2023-01-06. ATR 0.50 bottoms around 2023-09-13 at -17.84% closed-balance DD and does not regain the prior peak until 2024-05-23. ATR 0.25 bottoms later, around 2024-03-01 at -33.07%, and recovers only by 2024-07-11.

Practical implication: the 0.25 stop has a better edge but at 1% account risk it is aggressive for a strategy intended for tight funded-account drawdown rules. A lower risk percentage may be a more efficient solution than widening the stop if the goal is to preserve the 0.25 payoff geometry.

5. Trade Economics and Why ATR 0.25 Wins

Metric	ATR 0.25	ATR 0.50
Trades	1,697	1,697
Win / loss / BE	662 / 1,034 / 1	845 / 851 / 1
Win rate (independent)	39.01%	49.79%
Avg winning trade return	1.86%	0.83%
Avg losing trade return	-0.93%	-0.67%
Return payoff ratio	2.01x	1.25x
Break-even win rate	32.45%	44.31%
Actual win-rate cushion	6.56%	5.49%
Expected return / trade	0.163%	0.081%
Avg holding time	7.83 h	9.95 h
Avg winner holding time	10.89 h	10.95 h
Avg loser holding time	5.87 h	8.96 h



ATR 0.25 is a classic breakout distribution: frequent approximately 1R losses, fewer winners, but winners are much larger. ATR 0.50 raises the hit rate by giving trades more room, yet because risk-based sizing roughly halves position size and many surviving trades finish with modest EOD gains or losses, the average winner is much smaller.

6. Same-Signal Paired Comparison

Outcome on same entry date	Count	% of all trades	Avg ATR 0.25 return	Avg ATR 0.50 return
Both win	662	39.01%	1.86%	0.90%
0.25 loses / 0.50 wins	183	10.78%	-0.99%	+0.60%
Both lose	851	50.15%	-0.91%	-0.67%
Both break even	1	0.06%	0.00%	0.00%

The two variants are strongly related: trade-return correlation is 0.854, weekly-return correlation is 0.860, and monthly-return correlation is 0.863. They are therefore not separate portfolio systems; they are different risk/payoff expressions of the same edge.

A particularly important result is that there are 183 trades where ATR 0.25 is stopped out but ATR 0.50 later wins. The tighter stop knowingly gives up those recoveries. However, on the 662 trades where both variants win, ATR 0.25 averages about +1.86% versus +0.90% for ATR 0.50. The larger winners more than compensate for the additional stop-outs over the full sample.

ATR 0.50 actually produces the better return on 975 of 1,697 individual trades, versus 721 for ATR 0.25. Yet ATR 0.25 still finishes far ahead because the magnitude of its winning tail is much larger.

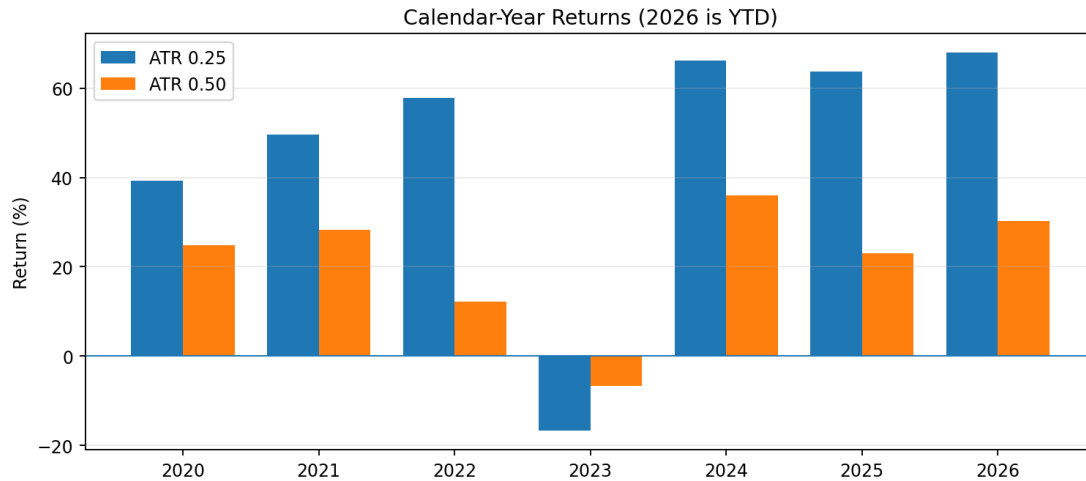
7. Exit-Reason Structure

Variant	Exit type	Trades	% trades	Win rate	Net profit	Avg P/L	Avg hold
ATR 0.25	SL	937	55.22%	0.00%	\$-333,743.74	\$-356.18	5.42 h
ATR 0.25	EOD/Other	760	44.78%	87.11%	\$449,018.21	\$590.81	10.80 h
ATR 0.50	SL	462	27.22%	0.00%	\$-82,898.85	\$-179.43	7.48 h
ATR 0.50	EOD/Other	1235	72.78%	68.42%	\$109,431.52	\$88.61	10.88 h

ATR 0.25 hits its stop on 55.2% of all trades, compared with only 27.2% for ATR 0.50. But when a 0.25 trade survives until EOD/other exit, 87.1% of those survivors are profitable and the average P/L is very large. This confirms that the tighter stop acts as an aggressive filter: it quickly rejects marginal breakouts and concentrates capital into the days that genuinely expand.

ATR 0.50 keeps substantially more trades alive, which explains its 49.8% overall hit rate and smoother drawdown. The trade-off is lower convexity: surviving moves are held with a smaller position because the 1% risk is spread across a stop twice as wide.

8. Year-by-Year Performance



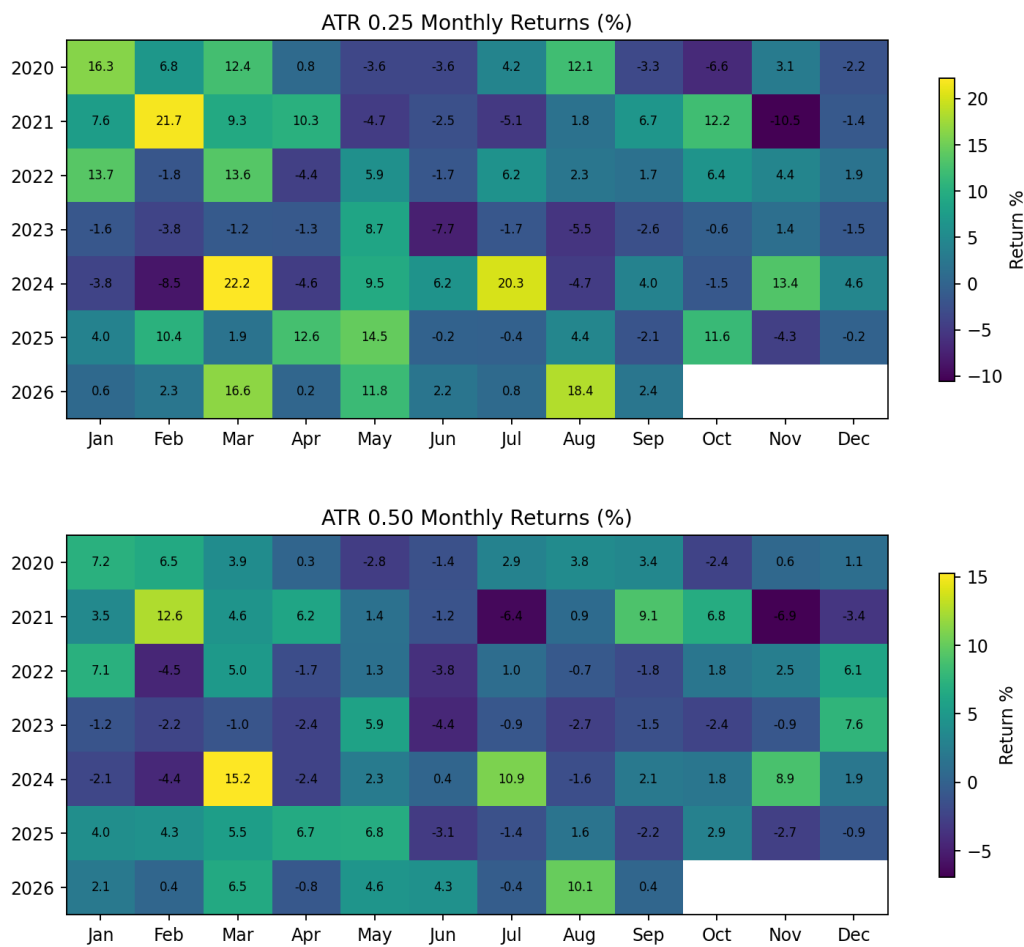
Year	ATR 0.25 Return	ATR 0.25 P/L	ATR 0.50 Return	ATR 0.50 P/L	Trades
2020	39.25%	\$3,925.40	24.88%	\$2,488.11	251
2021	49.66%	\$6,915.19	28.28%	\$3,531.06	252
2022	57.90%	\$12,066.73	12.23%	\$1,959.68	256
2023	-16.75%	\$-5,511.03	-6.67%	\$-1,199.27	254
2024	66.26%	\$18,153.58	35.95%	\$6,032.98	258
2025	63.75%	\$29,039.94	22.98%	\$5,241.95	254
2026 YTD	67.95%	\$50,684.66	30.22%	\$8,478.16	172

Both versions are positive in six of seven calendar-year buckets. 2023 is the only losing year: -16.75% for ATR 0.25 and -6.67% for ATR 0.50. ATR 0.25 then rebounds strongly in 2024, 2025 and 2026 YTD. This pattern supports the idea that the system is regime-sensitive rather than permanently stable.

2026 is incomplete and should not be compared directly with full calendar years.

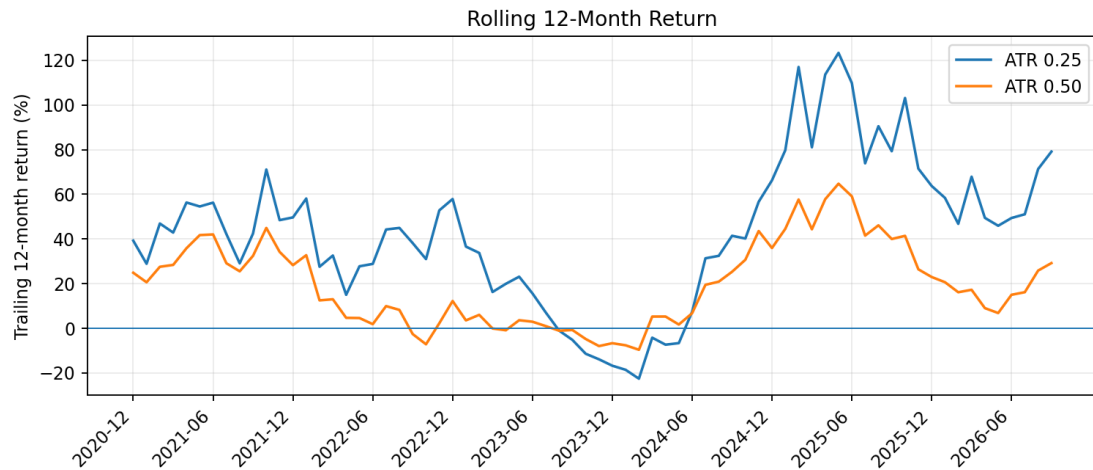
9. Monthly Performance

Metric	ATR 0.25	ATR 0.50
Months	81	81
Positive months	48 / 81 (59.26%)	48 / 81 (59.26%)
Average month	3.43%	1.71%
Median month	1.90%	1.13%
Monthly volatility	7.44%	4.41%
Best month	2024-03: 22.20%	2024-03: 15.25%
Worst month	2021-11: -10.52%	2021-11: -6.89%
Longest positive-month run	9 months	9 months
Longest negative-month run	5 months	6 months



Both variants have exactly the same number of positive and negative months (48 positive, 33 negative). The 0.25 advantage therefore comes mainly from **magnitude**, not a higher frequency of winning months.

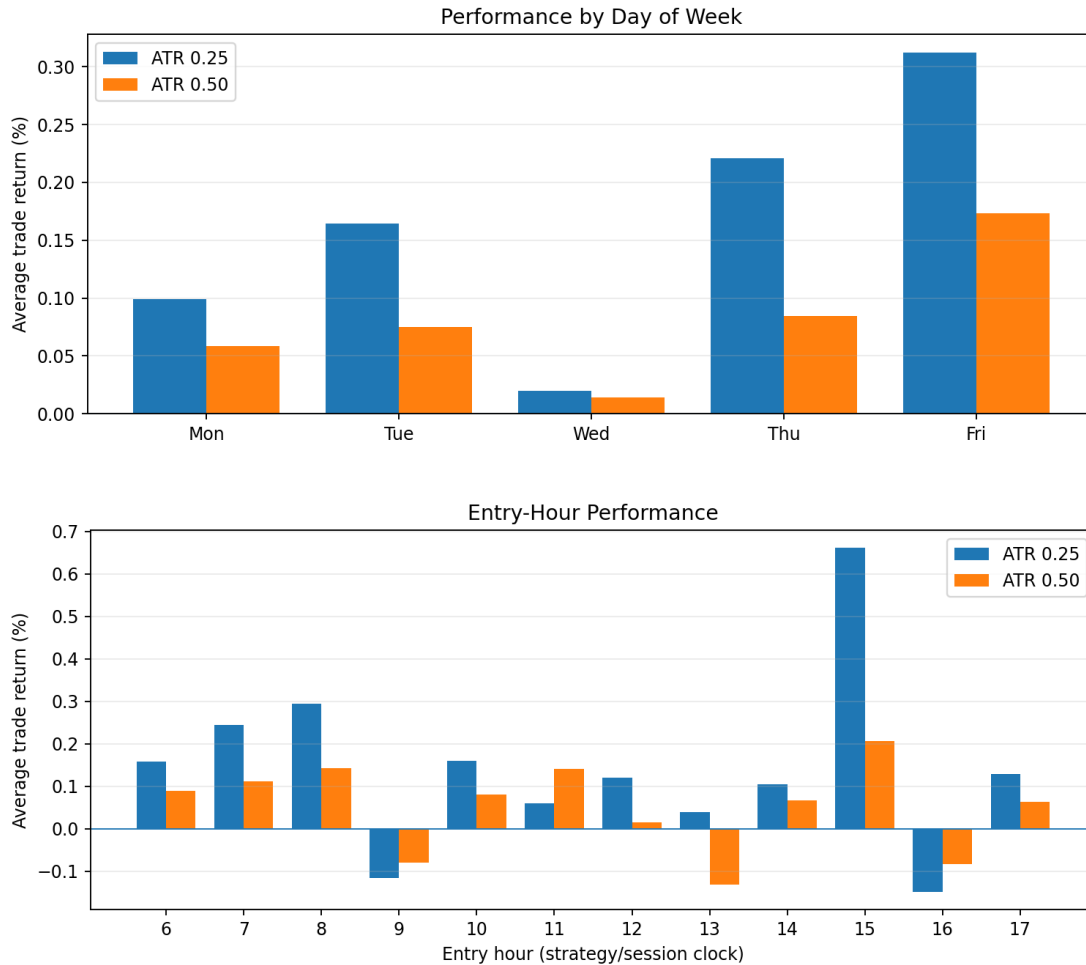
10. Weekly and Rolling 12-Month Behavior



Metric	ATR 0.25	ATR 0.50
Weeks	350	350
Positive weeks	54.57%	55.71%
Average week	0.79%	0.39%
Median week	0.43%	0.29%
Weekly volatility	3.77%	2.06%
Best week	14.73%	8.67%
Worst week	-5.01%	-4.71%

ATR 0.25 is more volatile, but its average week is approximately twice as strong. Rolling 12-month returns clearly expose the 2023-2024 weak regime and should be used as a robustness diagnostic during future optimization.

11. Timing Analysis



Day	Trades	ATR 0.25 avg ret	ATR 0.50 avg ret	ATR 0.25 win%	ATR 0.50 win%
Mon	334	0.099%	0.058%	39.22%	48.20%
Tue	346	0.164%	0.075%	41.33%	52.02%
Wed	341	0.020%	0.014%	35.19%	47.21%
Thu	341	0.221%	0.085%	38.12%	48.68%
Fri	335	0.312%	0.174%	41.19%	52.84%

Day-of-week: Friday is the strongest day in both variants, while Wednesday is the weakest. Treat this as an out-of-sample hypothesis, not an immediate filter.

Entry hour: the bulk of the edge occurs in entries from 06:00-08:59 in the strategy/session clock. The 09:00 bucket is negative in both variants across 185 trades. An entry-cutoff experiment is justified, but it must be validated on a holdout period.

12. Month-of-Year Seasonality

Month	0.25 Avg	0.25 Positive	0.50 Avg	0.50 Positive
Jan	5.26%	71.43%	2.95%	71.43%
Feb	3.87%	57.14%	1.83%	57.14%
Mar	10.70%	85.71%	5.67%	85.71%
Apr	1.95%	57.14%	0.83%	42.86%
May	6.01%	71.43%	2.78%	85.71%
Jun	-1.03%	28.57%	-1.31%	28.57%
Jul	3.47%	57.14%	0.82%	42.86%
Aug	4.14%	71.43%	1.62%	57.14%
Sep	0.98%	57.14%	1.34%	57.14%
Oct	3.59%	50.00%	1.42%	66.67%
Nov	1.25%	66.67%	0.25%	50.00%
Dec	0.20%	33.33%	2.06%	66.67%

March is the strongest recurring month for both configurations, while June is the clearest weak month: average June return is negative for both and only 2 of 7 Junes are positive. Seven observations per month are not enough for a production filter.

13. Profit Concentration and Tail Dependence

Top winning trades	ATR 0.25: share of total net profit	ATR 0.50: share of total net profit
1	5.37%	3.85%
5	24.81%	17.81%
10	41.12%	32.34%
20	64.58%	54.58%
50	117.90%	107.95%

For ATR 0.25, the top 10 winning trades contribute about 41.1% of total net profit; for ATR 0.50 the figure is about 32.3%. The top 50 winners exceed 100% of net profit in both cases, meaning the remaining trades collectively subtract value.

This is not automatically a weakness - it is the normal economic shape of many trend/breakout systems - but it means the strategy must be tested over long samples and across instruments. Missing a handful of major expansion days or filtering them out can destroy the edge.

Largest daily/trade outcomes:

Variant	Date	Side	Return	P/L	Exit
0.25	2023-10-13	BUY	8.75%	\$2,305.00	EOD/Other
0.25	2025-12-29	SELL	8.57%	\$5,832.80	EOD/Other
0.25	2021-08-06	SELL	8.33%	\$1,551.85	EOD/Other
0.25	2024-11-06	SELL	8.28%	\$3,137.47	EOD/Other
0.25	2020-01-08	SELL	8.09%	\$834.00	EOD/Other
0.50	2023-12-04	SELL	5.30%	\$839.86	EOD/Other
0.50	2023-10-13	BUY	4.15%	\$645.40	EOD/Other
0.50	2024-11-06	SELL	4.08%	\$832.39	EOD/Other
0.50	2020-01-08	SELL	4.07%	\$417.00	EOD/Other
0.50	2020-02-28	SELL	4.05%	\$444.70	EOD/Other

14. Directional Breakdown

Variant	Side	Trades	Win rate	Net profit	Avg P/L	Avg return
ATR 0.25	BUY	887	40.36%	\$65,689.60	\$74.06	0.164%
ATR 0.25	SELL	810	37.53%	\$49,584.87	\$61.22	0.162%
ATR 0.50	BUY	887	52.42%	\$19,970.29	\$22.51	0.111%
ATR 0.50	SELL	810	46.91%	\$6,562.38	\$8.10	0.048%

Both long and short signals are profitable. There is no evidence here that one side should simply be disabled; direction-specific filtering should only be considered if it improves out-of-sample robustness.

15. What to Test Next - Prioritized Research Plan

Priority	Experiment	Why it matters
1	Stop frontier: 0.25, 0.30, 0.35, 0.40, 0.50 ATR	Find a stable plateau between growth and drawdown; avoid optimizing only endpoints.
2	Risk scaling for 0.25: 0.50%, 0.75%, 1.00%	Test whether the superior payoff geometry survives with prop-friendly drawdown.
3	Daily-ATR opening-range filter	Current tests use no range filter. Test max range around 0.35, 0.50, 0.65, 0.80 Daily ATR.
4	Entry cutoff	06-08 entry hours carry most edge; 09:00 bucket is negative in both variants. Test a cutoff without ass
5	Transaction-cost sensitivity	Explicit commissions are zero in report. Add realistic commission and worse slippage/spread assumpt
6	Walk-forward / holdout split	Parameter choice must be made on one segment and validated on unseen years.
7	Monte Carlo trade-order resampling	Estimate plausible worst drawdown and loss streak beyond the single historical path.

Recommended immediate sequence: keep the signal logic unchanged. First map the stop-risk frontier. Then add only the Daily-ATR range filter. Do not simultaneously optimize stop, range filter, cutoff and session time, because you will lose the ability to identify which change actually improved the system.

For the range filter, use the previous completed D1 ATR(14) so today's threshold cannot move while the opening range is forming. Record both rejected and accepted days; the goal is to improve drawdown and expectancy without filtering out the few major trend days that create a large portion of profit.

16. Final Assessment

Are the tests good? Yes - they are promising and worth continued research. The sample is large, covers multiple market regimes, and both stop variants remain profitable over the same 1,697 signals. ATR 0.25 improves not only terminal profit but also profit factor, expectancy, independent CAGR, monthly Sharpe/Sortino and Calmar. This is a meaningful result.

Which version is better? If the objective is raw growth and the account can tolerate deep drawdowns, ATR 0.25 is clearly superior in this sample. If the objective is a smoother curve, smaller loss streaks and easier funded-account risk management, ATR 0.50 is easier to live with. The more interesting production candidate may be ATR 0.25 with lower account risk, or a stop in the 0.30-0.40 region.

What would make me reject the strategy? If the edge disappears after realistic commission/slippage, if the 0.25 advantage is isolated to a narrow stop value rather than a plateau, if the range filter only improves the in-sample period but hurts holdout years, or if Monte Carlo drawdowns exceed the account's risk tolerance.

Current research status: PROMISING, NOT YET PRODUCTION-FROZEN. The core ORB concept has enough evidence to justify focused refinement before moving to another trading system.

All derived calculations use the completed deals in the two uploaded Strategy Tester workbooks. Daily figures are trade-day figures because the EA is capped at one trade per day. Weekly periods use ISO weeks; monthly and annual returns are compounded from actual start/end closed balances. CAGR uses the full test span from 2020-01-01 through the final completed trade on 2026-09-08.